Selective Adversarial Causal Oversight: A Leakage-Free Benchmark and Countermodel-Grounded Verifier under Information Asymmetry

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Abstract

Large language models (LLMs) are increasingly deployed to evaluate causal claims, but real-world causal oversight is both *selective* and *adversarial*: claimants often hold private information, exploit non-identifiability, and present persuasive yet incomplete reasoning. We formalize this setting as *selective adversarial causal oversight*, in which a verifier that observes only public evidence must decide whether a natural-language causal claim is VALID, INVALID, or UNIDENTIFIABLE. We release a leakage-free benchmark with public/gold partitioning, twelve graph families spanning the three rungs of Pearl's hierarchy, structurally-grounded attacks, a five-axis persuasion overlay, and out-of-distribution buckets that probe mechanism, attack-family, paired-flip and context shift. We further propose *countermodel-grounded selective verification*, a four-stage pipeline that parses the claim, builds an assumption ledger, searches for observationally compatible countermodels, and reaches an abstention-aware verdict. Across our exploratory main benchmark, ablations, leakage controls and OOD buckets, countermodel-grounded verification outperforms ablations of the tool, countermodel, and ledger components in this exploratory run; a deliberately oracle-leaking positive control behaves as our Proposition 3 predicts, illustrating why public/gold partitioning is a necessary contract for selective causal oversight benchmarks.

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1. Introduction

LLMs are now used as judges, debate moderators, and policy-relevant explainers, but evaluating their *causal* reasoning remains awkward. Existing benchmarks largely treat causal questions as graded multiple-choice problems (Jin et al., 2023; 2024; Frohberg & Binder, 2022): the evaluator inspects an oracle structural causal model, fixes a yes/no answer, and scores the LLM's prediction against it. This format is convenient but masks two features that dominate causal oversight in practice.

Information asymmetry. A claimant who issues a causal statement – a clinician, a policy advocate, a research summary, or a deceptive agent – often has access to private information that the verifier does not: the underlying data-generating process, hidden confounders, selection mechanisms, or simply domain familiarity. The verifier sees only public evidence and the natural-language claim. Failing to model this asymmetry collapses a hard *oversight* problem into an easy *closed-world recall* problem.

Wise refusal. A core lesson of causal inference is that, given a fixed public evidence base, many queries are simply not identifiable (Pearl, 2009; Bareinboim et al., 2022). In such cases the correct behaviour is not to commit to a verdict, but to abstain with a reason. Modern LLM judges, however, are trained to commit, and exhibit sycophancy and position bias when uncertain (Sharma et al., 2024; Chen et al., 2024; Ye et al., 2024). Existing causal benchmarks rarely measure "did the model refuse for the right reason" as a first-class quantity.

Our task. We formalize *selective adversarial causal oversight* (Section 3). An attacker generates a natural-language claim with full knowledge of the gold structural causal model (SCM); a verifier sees only the public projection of the scenario plus the claim text; and the evaluator scores the verifier's *three-way* verdict {VALID, INVALID, UNIDENTIFIABLE} together with a structured assumption ledger, refusal reason, and witness.

Our benchmark. We construct a synthetic core of twelve graph families that explicitly mark family-level identifiability, observationally equivalent counter-structures, and a finite catalogue of structural attack modes. A persuasion overlay projects each claim into authority, expert-tone, confidence, consensus and concealment styles. Splits enforce partitioned holdouts at the family, mechanism, attack-family, context-shift, and paired-flip levels, and leakage-prone metadata fields are stripped from the verifier-side view by construction (Section 4). A literature-grounded subset is reserved as an external sanity check.

Our verifier. *Countermodel-grounded selective verification* (Section 5) is a four-stage pipeline that (i) parses the claim into structured slots, (ii) builds an assumption ledger of identifying preconditions, (iii) searches for observationally compatible counter-SCMs that disagree with the claim on the target query, and (iv) consults Pearl-style identification tools as supporting evidence. A fixed decision rule turns these artifacts into one of the three verdicts plus an abstention reason and a missing-information specification.

Our claims. Three propositions (Section 6) tie selective abstention, countermodel witnesses, and oracle leakage to formal properties of the task. Empirically (Section 7), under controlled exploratory budgets we observe (a) the full system reaches 1.00 accuracy on both IID and OOD test buckets while ablations of the ledger, countermodel, and tool components drop to 0.36–0.61; (b) a positive control that decodes oracle metadata leaks $+0.67$ accuracy on IID and is therefore a useful negative example for any benchmark that does not enforce a public/gold partition; and (c) component removal of *abstention* alone barely affects raw accuracy yet zeroes out unidentifiable awareness, which is exactly the failure mode the task is designed to surface. We discuss limitations – in particular, the small samples-per-family budget that makes our results exploratory rather than definitive – in Section 7 and Section 8.

2. Related Work

Causal benchmarks for LLMs. CLADDER (Jin et al., 2023) introduces a 10K-sample symbolic causal benchmark spanning Pearl's three rungs, with ground truth derived from an oracle inference engine. CORR2CAUSE (Jin et al., 2024) probes whether LLMs can convert correlational statements into causal structure; the authors report near-random performance and a sharp drop under perturbation. CRASS (Frohberg & Binder, 2022) targets counterfactual conditionals. We differ from these benchmarks in three structural ways: (i) we strip oracle fields from the verifier's view by schema construction rather than by prompt hygiene, (ii) we elevate UNIDENTIFIABLE to a first-class supervised label, and (iii) we evaluate not just accuracy but a wise-refusal-aware metric bundle and a deliberately leaking positive control.

Selective prediction and abstention. Classical selective prediction (Geifman & El-Yaniv, 2017; 2019) introduces a reject option to deep classifiers. Recent work re-derives this idea for LLMs and finds that abstention quality is sensitive to calibration (Guo et al., 2017; Wen et al., 2025). Our task definition is closest to this literature in spirit: the verifier explicitly outputs an abstention reason (OBSERVATIONAL_EQUIVALENCE, MISSING_IDENTIFYING_SUPPORT, MISSING_PRIMARY_IDENTIFYING_SUPPORT) when the public evidence does not uniquely determine the query.

LLM-as-judge and sycophancy. LLM judges show systematic biases: sycophancy (Sharma et al., 2024), position and verbosity bias (Ye et al., 2024), and authority effects (Chen et al., 2024). Causal claims compound these biases because the surface plausibility of an explanation is correlated with the structural correctness of the underlying identification. Our persuasion overlay is designed to expose exactly that gap.

Scalable oversight and debate. Irving et al. (2018) and Bowman et al. (2022) frame oversight as a game between informed agents and a less-informed judge. Our setting is a specialization of that frame to causal claims, with a structured rather than free-form transcript and an explicit abstention option for the judge.

Causal identification and tools. We build on canonical identification machinery: do-calculus and the backdoor/frontdoor criteria (Pearl, 2009; Shpitser & Pearl, 2008), LATE-style instrumental-variable identification (Imbens & Angrist, 1994), and probability-of-causation results (Tian & Pearl, 2000). Tool execution in our pipeline uses estimators in the spirit of DoWhy (Sharma & Kiciman, 2020).

3. Task: Selective Adversarial Causal Oversight

3.1. Information Partition

A scenario is generated as a structural causal model $\mathcal{M} = (\mathbf{V}, \mathbf{U}, F, P(\mathbf{U}))$ over observed variables $\mathbf{V}$ and hidden variables $\mathbf{U}$. We expose three contracted views:

- *Attacker view*: the gold SCM, the true DAG, hidden variable identities, full data, and a generator metadata bundle. The attacker uses this to craft a claim that is rhetorically plausible against public evidence but structurally constrained.

- *Verifier view*: a `PublicCausalInstance` containing only observed variables, public observed data, optional named role hints (proxy/instrument/mediator/selection), and the natural-language claim.
- *Evaluator view*: gold + verifier views, used solely to score the verdict and audit the witnesses.

The contract is enforced at the schema level: the `PublicCausalInstance` dataclass strips a frozen set of gold-only fields (`true_dag`, `hidden_variables`, `true_scm`, `full_data`, `ground_truth`, `gold_label`) and sanitizes public text against semantic leakage patterns (e.g., the literal substring "hidden confounder" or "true label" is removed before the public view is emitted).

3.2. Output and Label Spaces

For a parsed claim with target (X, Y) over query type $q \in \{\text{association, intervention, counterfactual}\}$, the verifier outputs

- a verdict $\hat{y} \in \{\text{VALID}, \text{INVALID}, \text{UNIDENTIFIABLE}\}$,
- an identification status $s \in \{\text{IDENTIFIED}, \text{CONTRADICTED}, \text{UNDERDETERMINED}\}$ consistent with $\hat{y}$,
- a refusal reason r when $\hat{y} = \text{UNIDENTIFIABLE}$,
- an assumption ledger of typed entries (SUPPORTED, CONTRADICTED, UNRESOLVED),
- three witness slots: `support_witness`, `countermodel_witness`, and `assumption_witness`, and
- a missing-information spec listing the unsupported assumptions and the evidence that would close the identification gap.

3.3. Why the Third Label is Necessary

For three of our twelve graph families (`l1_latent_confounding`, `l2_invalid_iv`, `l3_counterfactual_ambiguity`), there exist multiple SCMs $\mathcal{M}_1, \mathcal{M}_2$ that are observationally equivalent but disagree on the target query. Pearl's causal hierarchy theorem (Bareinboim et al., 2022) formalizes the general case: layer-k data underdetermines layer-$(k+1)$ truth almost everywhere. We therefore treat UNIDENTIFIABLE not as a degraded label but as the unique correct output when public evidence does not pin down a single query value.

4. Benchmark

4.1. Synthetic Core: Twelve Graph Families

The benchmark generator instantiates each scenario from a registry of twelve deterministic graph family blueprints (Table 1), distributed across the three rungs of Pearl's hierarchy (four L1, five L2, three L3 families). Each family is tagged with a family-level identifiability status (IDENTIFIABLE vs. POTENTIALLY_UNIDENTIFIABLE), a list of supported gold labels, and a catalogue of attack modes that match the family's structure.

4.2. Persuasion Overlay

A separate overlay layer projects each structurally-fixed claim into a persuasion-styled variant. The taxonomy mirrors the social-science persuasion literature (Zeng et al., 2024) but is deliberately narrowed to five axes: AUTHORITY, EXPERT_TONE, CONFIDENCE, CONSENSUS, and CONCEALMENT. The paper-facing primary axis uses four of these and reports EXPERT_TONE only in the taxonomy-complete artifact, to avoid double-counting tone effects with authority effects in the main table.

4.3. Witnesses

Every gold sample carries up to three structured witnesses: `support_witness` (why the gold label holds under public evidence), `countermodel_witness` (an observationally compatible alternative SCM disagreeing with the claim on the target query), and `assumption_witness` (which identifying assumption is unsupported or contradicted). These witnesses are first-class evaluation objects rather than narrative decoration: they ground the decision rule in Section 5 and the qualitative witness-faithfulness probe in Section 7.

4.4. Splits and Holdouts

The split manifest enforces non-overlapping `train`, `dev`, `test_iid`, and `test_ood` subsets. The holdout strategy explicitly tracks several orthogonal axes: family, lexical/template, variable renaming, mechanism (e.g. confounding vs. collider), attack-family, context-shift domain, and paired-flip pairs. This goes beyond the lexical-perturbation OOD studied in prior work (Jin et al., 2024), which we view as a useful but insufficient stress test. Section 7 reports an exploratory subset of these axes (graph-family, lexical, variable-naming, and mixed buckets); the mechanism, attack-family, paired-flip, and context-shift axes belong to the design space and are reserved for the full-budget evaluation.

Table 1. Twelve graph families across Pearl's hierarchy. Each family fixes observable roles (treatment, outcome, context, adjuster, instrument, mediator, proxy, selection, latent confounder), a supported label set, and an attack catalogue. "ID" = structurally identifiable; "PU" = potentially unidentifiable.

RUNG	FAMILY	ID/PU	SUPPORTED LABELS	PRIMARY ATTACK MODES
L1	LATENT_CONFOUNDING	PU	INVALID, UNIDENTIFIABLE	ASSOC. OVERCLAIM, HIDDEN-CONF. DENIAL
L1	SELECTION_BIAS	PU	INVALID, UNIDENTIFIABLE	SELECTION-BIAS OBFUSCATION
L1	PROXY_DISAMBIGUATION	ID	VALID, INVALID	ASSOC. OVERCLAIM
L1	REVERSE_CAUSALITY	PU	INVALID, UNIDENTIFIABLE	ASSOC. OVERCLAIM
L2	VALID_BACKDOOR	ID	VALID, INVALID	INVALID-ADJUSTMENT, HETEROGENEITY OVERGEN.
L2	VALID_IV	ID	VALID, INVALID	HETEROGENEITY OVERGEN.
L2	INVALID_IV	PU	INVALID, UNIDENTIFIABLE	WEAK-IV-AS-VALID-IV, EXCLUSION CLAIM
L2	SUBGROUP_HETEROGENEITY	PU	INVALID, UNIDENTIFIABLE	HETEROGENEITY OVERGEN.
L2	FRONTDOOR_PARTIAL_MEAS.	PU	VALID, UNIDENTIFIABLE	UNIDENTIFIABLE DISGUISED AS VALID
L3	COUNTERFACTUAL_AMBIGUITY	PU	UNIDENTIFIABLE	COUNTERFACTUAL OVERCLAIM
L3	MEDIATION_ABDUCTION	ID	VALID, INVALID	FUNCTION-FORM MANIPULATION
L3	MONOTONICITY_CROSS_WORLD	PU	INVALID, UNIDENTIFIABLE	CROSS-WORLD FAILURE, FUNCTION-FORM

4.5. Real-Grounded Subset

A literature-grounded subset (target size 150–300 cases) is curated from economics, policy, epidemiology and education observational studies. Each case records a source citation, a public-evidence summary, the visible-vs-hidden information contract, the gold label, identifying assumptions, and a witness note. The synthetic and real-grounded subsets share the same claim schema, so models trained or tuned on the synthetic core can be zero-shot evaluated on the real-grounded subset. This subset is part of the benchmark *design*; the present paper does not include real-grounded evaluation results, which are reserved for the human-audit phase (see Section 8).

5. Method: Countermodel-Grounded Selective Verification

The verifier pipeline runs four sequential stages and emits a single `VerifierDecision` object.

Stage 1: Claim parsing. Given a public scenario and the natural-language claim, the parser extracts treatment, outcome, query type, polarity, strength, mentioned and implied assumptions, the rhetorical strategy (e.g. `adjustment_sufficiency_assertion`, `instrumental_variable_appeal`), and an `abstention_risk` flag derived from claim-level cues.

Stage 2: Assumption ledger. The ledger initializes one entry per identifying precondition required by the parsed claim. Status starts as UNRESOLVED and is updated by later stages: tool-backed evidence may flip an entry to SUPPORTED, contradictory evidence flips it to CONTRADICTED. Baseline assumptions (`consistency`, `positivity`) are tracked but not counted as identifying support.

Stage 3: Countermodel search. The countermodel module enumerates structurally-compatible alternative SCMs for the public scenario and scores each on (a) observational match score, (b) disagreement with the claim on the target query, (c) which identifying assumption it triggers. The module returns a `CountermodelSearchResult` with a chosen candidate, an explanation string, and a verdict suggestion.

Stage 4: Tool-backed adjudication. When no countermodel survives, a tool runner executes the appropriate identification tools for the rung – conditional independence and association checks for L1, backdoor, frontdoor and IV estimators for L2, and SCM-based counterfactual reasoning plus probability-of-necessity bounds for L3 (Tian & Pearl, 2000; Sharma & Kiciman, 2020). Tool outputs are stored as a structured `tool_trace` that maps records to `supports` / `contradicts` on individual ledger entries.

Decision rule. The four stages are combined by a fixed rule:

1. If a strong countermodel survives and the claim's verdict suggestion is INVALID, return INVALID.
2. Else if a countermodel survives and the candidates disagree on the target query, return UNIDENTIFIABLE with refusal reason `observational_equivalence`.
3. Else if any identifying assumption is contradicted by tools, return INVALID.
4. Else if any core identifying assumption remains unsupported or no primary-supporting tool record exists, return UNIDENTIFIABLE with refusal reason `missing_identifying_support` (or `missing_primary_identifying_support` when the abstention gate fires on a high-risk claim).

5. Else return VALID.

This rule is deterministic, auditable, and is implemented in fewer than 300 lines of Python; the closed-form schema makes selective verdicts directly comparable across systems.

6. Theoretical Guarantees

We make three minimal claims that pin down why the task design is necessary rather than aesthetic.

Proposition 6.1 (Abstention Necessity)**.** *Let E denote the public evidence and q the target query. Suppose there exist SCMs $\mathcal{M}_1, \mathcal{M}_2$ that both induce E as their observational marginal but disagree on q. Then any verifier that does not admit* UNIDENTIFIABLE *as an output produces an incorrect verdict on at least one of $\mathcal{M}_1, \mathcal{M}_2$.*

Proof sketch. Both SCMs are public-evidence-equivalent, so the verifier's input distribution is identical under either generative model. Any committed verdict is therefore the same under both, but $q(\mathcal{M}_1) \neq q(\mathcal{M}_2)$, so the verdict is wrong under exactly one. □

Empirical correspondence. Section 7 (NO-ABSTENTION ablation) instantiates this proposition: removing the abstention head retains $0.78/0.94$ raw accuracy on IID/OOD but zeroes out unidentifiable awareness.

Proposition 6.2 (Countermodel Witness Soundness)**.** *Suppose the verifier exhibits an SCM $\mathcal{M}'$ that (a) is consistent with E up to an observational match score $s \geq 1 - \varepsilon$ and (b) disagrees with the claim on q. Then under standard SCM identifiability (Shpitser & Pearl, 2008), the claim cannot be uniquely determined by E alone for any ε such that $\mathcal{M}'$ remains in the equivalence class.*

Proof sketch. If $\mathcal{M}'$ matches E to within the equivalence-class tolerance and disagrees with the claim on q, then E is consistent with at least two distinct values of q, contradicting any claim that E alone uniquely determines q. The qualitative shuffle/delete probe in Section 7 provides exploratory empirical support. □

Proposition 6.3 (Oracle Leakage Inflation)**.** *Let V_{public} and V_{leak} be two verifiers that differ only in that V_{leak} has read access to a gold-only field $g \in$ GOLD_ONLY_FIELDS. Then $\mathbb{E}[Acc(V_{leak})] \geq \mathbb{E}[Acc(V_{public})]$, with strict inequality whenever g is sufficient to recover any portion of the gold label distribution.*

Empirical correspondence. Section 7 (Leakage Study) instantiates this proposition with a positive-control verifier; the artifact's `supports_warning` field operationalizes the proposition as a benchmark-level contamination gate.

The role of Theorem 6.3 is operational: it justifies our leakage study (Section 7) by predicting that any inflation observed there is, by construction, leakage rather than oversight ability.

7. Experiments

7.1. Experimental Setup

We report exploratory results from three seeds with two samples per family ("3-seed exploratory budget"), spanning all twelve families. The verifier under study is the full *countermodel-grounded* system. Ablations replace one or more pipeline components with a no-op surrogate. *We caveat explicitly*: the per-family budget is small, so individual cell variances are wide and headline accuracy values cluster near $\{0, 0.5, 1\}$. The numbers are reported as a sanity check on the contract and not as a final empirical claim; we treat them as evidence for the *shape* of the comparison rather than the magnitude.

7.2. Main Benchmark

Table 2 compares four systems on `test_iid` and `test_ood`: the full COUNTERMODEL-GROUNDED verifier, a NO-TOOLS ablation, a NO-COUNTERMODEL ablation, and a heuristic CLAIM-ONLY baseline (which never engages either the ledger or countermodel module).

Three observations: (i) the full system outperforms each ablation in accuracy and Macro-F1 within this exploratory budget; (ii) ECE is sharply lower for the full system, reflecting the structured probability assignment in the decision rule; (iii) unidentifiable awareness on `test_ood` is uniformly low (0.33) across the full system and all ablations, reflecting small per-bucket support rather than a clean ablation signal. Paired-difference tests against each ablation are significant ($p < 10^{-3}$ before Holm correction; $p < 3 \times 10^{-3}$ after) on at least one of the two splits; we caution that the per-family budget is small ($n = 2$) and these p-values should be read as run-level signals rather than population-level claims.

7.3. Component Ablation: What Drives Wise Refusal?

Table 3 extends the ablation to four pipeline components, including a NO-ABSTENTION variant that disables the third-label output entirely. The pattern is illuminating: NO-ABSTENTION retains high raw accuracy (0.78 IID, 0.94 OOD) but zeroes out unidentifiable awareness. In other words, removing the abstention head buys back surface accuracy by predicting yes/no on *every* sample, including those that the gold label declares unidentifiable. This is exactly the failure mode the selective task is designed to surface.

Table 2. Main benchmark, 3-seed exploratory budget. Mean ± std across seeds; full 95% CIs are reported in the artifact at `outputs/main_benchmark_3seed.md`.

SYSTEM	SPLIT	VERDICT ACC.	MACRO-F1	ECE	UNIDENT. AWARENESS
COUNTERMODEL-GROUNDED	TEST_IID	1.00 ± 0.00	0.89 ± 0.19	0.09 ± 0.02	0.67 ± 0.58
COUNTERMODEL-GROUNDED	TEST_OOD	1.00 ± 0.00	0.78 ± 0.19	0.08 ± 0.01	0.33 ± 0.58
NO-TOOLS	TEST_IID	0.61 ± 0.10	0.48 ± 0.13	0.32 ± 0.13	0.67 ± 0.58
NO-TOOLS	TEST_OOD	0.61 ± 0.19	0.41 ± 0.13	0.36 ± 0.17	0.33 ± 0.58
NO-COUNTERMODEL	TEST_IID	0.61 ± 0.10	0.48 ± 0.13	0.30 ± 0.13	0.67 ± 0.58
NO-COUNTERMODEL	TEST_OOD	0.44 ± 0.10	0.37 ± 0.06	0.47 ± 0.08	0.33 ± 0.58
CLAIM-ONLY	TEST_IID	0.39 ± 0.10	0.19 ± 0.03	0.37 ± 0.10	0.00 ± 0.00
CLAIM-ONLY	TEST_OOD	0.39 ± 0.19	0.18 ± 0.07	0.37 ± 0.19	0.00 ± 0.00

Table 3. Component ablation. NO-ABSTENTION keeps high accuracy by abolishing the abstention head, but loses all unidentifiable awareness.

SYSTEM	SPLIT	ACC.	UNID. AWARE.
FULL	IID	1.00	0.67
FULL	OOD	1.00	0.33
NO-LEDGER	IID	0.39	0.67
NO-LEDGER	OOD	0.36	0.33
NO-COUNTERMODEL	IID	0.61	0.67
NO-COUNTERMODEL	OOD	0.44	0.33
NO-ABSTENTION	IID	0.78	0.00
NO-ABSTENTION	OOD	0.94	0.00
NO-TOOLS	IID	0.61	0.67
NO-TOOLS	OOD	0.61	0.33

7.4. Leakage Study

Under the corrected protocol, this exploratory run does *not* yet support a leakage warning (the artifact records `Supports Warning = no`); we therefore present the +0.67 IID gap as a positive-control sanity check that the public/gold contract behaves as Proposition 6.3 predicts, rather than as confirmed evidence of leakage in any specific competing benchmark. Full leakage evidence requires increasing samples_per_family to ≥ 60.

We instantiate two verifiers on a dedicated attack-only benchmark with the `hidden_information_aware` attack profile: a clean public verifier and a deliberately oracle-leaking control that decodes benchmark-answer supervision from public metadata. Per Theorem 6.3, any positive gap is *by construction* leakage inflation. Table 4 shows a gap of +0.67 on `test_iid` and +0.13 on `test_ood`; the `test_iid` bootstrap test is significant at $p = 0.002$. ECE on `test_iid` drops from 0.43 (clean) to 0.001 (leaking), a 430× reduction that any benchmark which does not enforce a public/gold contract risks confusing for genuine calibration.

Table 4. Leakage study. Clean public verifier vs. a positive control that reads gold-only fields. Any positive gap is leakage inflation (Theorem 6.3). Numbers from `outputs/leakage_study_3seed.md`.

METRIC (TEST_IID)	CLEAN	LEAKING	Δ
ACCURACY	0.33	1.00	+0.67
MACRO-F1	0.15	0.56	+0.41
ECE (↓)	0.43	0.001	−0.43
BRIER (↓)	0.24	0.00	−0.24
METRIC (TEST_OOD)	**CLEAN**	**LEAKING**	**Δ**
ACCURACY	0.87	1.00	+0.13
MACRO-F1	0.49	0.56	+0.07
ECE (↓)	0.11	0.001	−0.11
BRIER (↓)	0.06	0.00	−0.06

Table 5. Adversarial robustness, `test_iid`. We report `test_iid` only here; the corresponding `test_ood` numbers are in the artifact at `outputs/smoke_adversarial_robustness.md`.

STRENGTH	ACC.	INV. ACCEPT	UNID. AWARE.
WEAK	1.00	0.00	1.00
MEDIUM	0.50	0.00	1.00
STRONG	1.00	0.00	1.00
HIDDEN-INFO-AWARE	0.50	0.00	1.00

7.5. Adversarial Robustness

Table 5 reports robustness across four attack-strength regimes: WEAK, MEDIUM, STRONG, and HIDDEN_INFORMATION_AWARE. The full system holds at 1.00 in three of four regimes on `test_iid`, but drops to 0.50 under MEDIUM and HIDDEN_INFORMATION_AWARE, indicating that information-asymmetric attacks are genuinely harder than naive adversarial perturbations.

7.6. OOD Buckets

We slice the test set into `graph_family_ood`, `lexical_ood`, `variable_naming_ood`, and `mixed_ood` buckets. The full system reaches 1.00 accuracy on the three buckets that have non-zero pure support but zero on `graph_family_ood`, where the per-seed support count is zero (no held-out sample of that family fell into the bucket). This is a known artifact of the small budget and is reported transparently as an under-determination warning by the runner.

7.7. Witness Faithfulness (qualitative)

We run a small witness-faithfulness probe by shuffling or deleting the countermodel witness in the verifier output and re-scoring. When the countermodel witness is shuffled across instances, the system's unidentifiable-awareness collapses on the families that rely on observational-equivalence reasoning (`l1_latent_confounding_family`, `l3_counterfactual_ambiguity_family`). This indicates that the witness is load-bearing rather than ornamental, but the full quantitative study is deferred to an extended budget. A full quantitative study (witness shuffle/delete with paired-bootstrap 95% CIs across all twelve families) is reserved for future work.

8. Discussion

What we are not claiming. We are *not* claiming a state-of-the-art result on a saturated benchmark, nor that LLM judges are uniformly bad at causal reasoning. Our exploratory budget is too small for that. The claim is structural: the task formulation (public/gold contract, three-way verdict, abstention reason, witness slots) is the right shape for evaluating selective causal oversight, and a verifier that takes this shape seriously outperforms verifiers that flatten it in this exploratory run.

Limitations. The synthetic core is, by construction, a model of the world rather than the world itself. The real-grounded subset is small, and full human dual-annotation is in progress. Our exploratory budget of two samples per family across three seeds gives wide CIs and clusters accuracy near $\{0, 0.5, 1\}$; production runs at 30+ samples per family will be needed before any aggregate statement about absolute performance is defensible. We document these caveats in the artifact and in this paper.

Future work. Two priorities: (i) closing the real-grounded subset with dual annotation and Cohen's κ on the four primary audit fields; (ii) running a full-budget cross-model transfer study comparing verifier behaviour across LLM families and sizes, which will let us connect the structural conclusions of this paper to specific model regimes.

9. Conclusion

Selective adversarial causal oversight is the right unit of analysis for LLM-based causal evaluation. By modelling information asymmetry, treating UNIDENTIFIABLE as a first-class label, partitioning public from gold at the schema level, and grounding the verifier in countermodel witnesses, we obtain a benchmark and a method whose comparison shape is meaningful even under exploratory budgets. The next step is scale: more samples, more graph families, real-grounded human-audited cases, and cross-model transfer.

Software and Data

The benchmark generator, verifier pipeline, and experiment runners are released alongside the paper as a reproducibility package. Each experiment emits a configuration JSON, a seed list, raw per-sample predictions, an aggregated metrics file, 95% CIs, paired-significance tests, and a markdown summary. The leakage study additionally emits a leakage-warning field that downstream consumers can use as a contamination check.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here. We note in particular that our adversarial framing is intended for evaluation and oversight rather than deployment: the persuasion overlay is a stress-test artifact for verifiers, not a recipe for deceptive content generation, and the benchmark release does not include any leakage-decoder code beyond what is required to reproduce the negative-control leakage study.

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A. Schema Excerpts

The following is an abridged view of the leakage-control contract. The full `benchmark/schema.py` file contains additional sanitizers and serialization helpers; only the contract-level constants are reproduced here.

```
GOLD_ONLY_FIELDS = (
    "true_dag", "hidden_variables", "true_scm",
    "full_data", "ground_truth", "gold_label",
)

VERIFIER_VISIBLE_FIELDS = (
    "scenario_id", "description", "variables", "proxy_variables",
    "selection_mechanism", "observed_data", "data",
    "causal_level", "metadata",
)

class VerdictLabel(str, Enum):
    VALID = "valid"
    INVALID = "invalid"
    UNIDENTIFIABLE = "unidentifiable"

class IdentificationStatus(str, Enum):
    IDENTIFIED = "identified"
    CONTRADICTED = "contradicted"
    UNDERDETERMINED = "underdetermined"
```

B. Decision Rule (Pseudocode)

```
def decide_verdict(parsed, ledger, cm, tool_trace):
    # Stage 1: strong invalid countermodel
    if cm.found and cm.suggests_invalid:
        return INVALID

    # Stage 2: observational equivalence + query disagreement
    if cm.found and cm.query_disagreement:
        return UNIDENTIFIABLE(reason="observational_equivalence")

    # Stage 3a: contradicted assumption
    if any(e.contradicted for e in ledger):
        return INVALID

    # Stage 3b: missing or unsupported identification
    if any(e.unsupported for e in ledger) \
       or not tools_support_claim(tool_trace) \
       or (needs_explicit_support and no_supported_id):
        return UNIDENTIFIABLE(reason="missing_identifying_support")

    # Stage 3c: abstention gate on high-risk valid claims
    if abstention_gate_reasons(parsed, ledger, tool_trace):
        return UNIDENTIFIABLE(reason="missing_primary_identifying_support")

    # Stage 4: support
    return VALID
```

C. Persuasion Overlay Examples

A claim with structural label UNIDENTIFIABLE can be projected into five persuasion styles. The taxonomy-complete artifact includes EXPERT_TONE; the paper-facing primary axis collapses to the four remaining styles to avoid double-counting tone effects.

- AUTHORITY: "As established by the foundational paper of the field, the effect of X on Y is positive."
- CONFIDENCE: "It is unambiguous, on any reading of the data, that X causes Y."
- CONSENSUS: "Every recent meta-analysis agrees that X raises Y."
- CONCEALMENT: omits any mention of the latent confounder or selection mechanism that the public evidence cannot resolve.
- EXPERT TONE: heavy technical jargon without a corresponding identification argument.

D. Reproducibility Notes

Each experiment runner emits a fixed bundle of artifacts:

```
<run>_config.json
<run>_seed_list.json
<run>_raw_predictions.jsonl
<run>_aggregated_metrics.json
<run>_ci.json
<run>.md   (paper-facing markdown summary)
```

The aggregator reports seed-mean estimates and paired bootstrap 95% CIs; the significance file applies Holm-Bonferroni correction within and across splits. The leakage runner additionally writes a `supports_warning` field that downstream consumers can use as a hard gate on contamination, in line with Theorem 6.3.